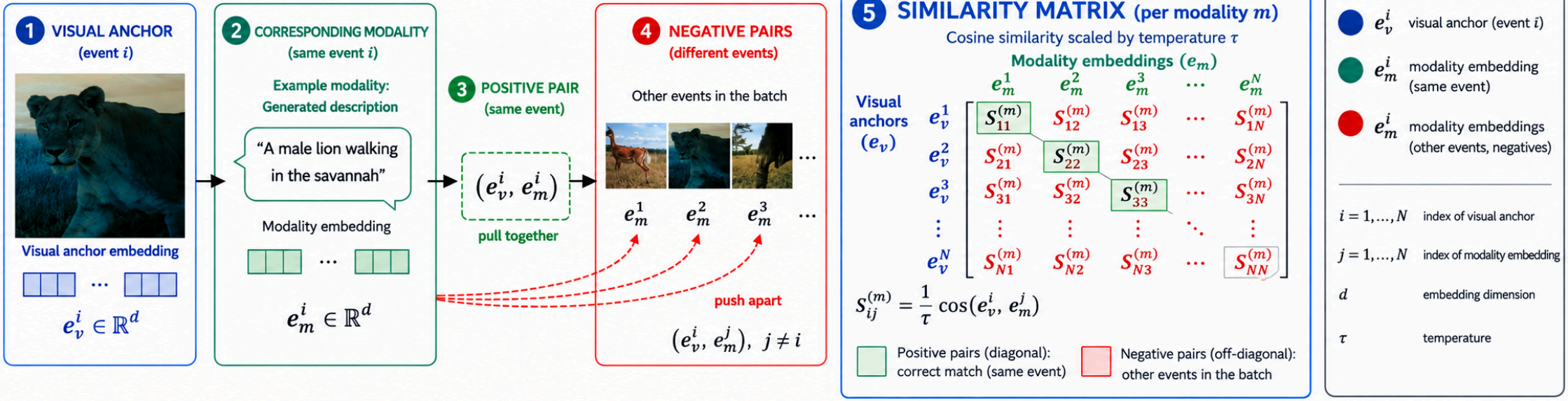

1 VISUAL ANCHOR
(event i)
Visual anchor embedding
$e_v^i \in \mathbb{R}^d$
2 CORRESPONDING MODALITY
(same event i)
Example modality:
Generated description
"A male lion walking in the savannah"
Modality embedding
$e_m^i \in \mathbb{R}^d$
3 POSITIVE PAIR
(same event)
(e_v^i, e_m^i)
pull together
4 NEGATIVE PAIRS
(different events)
Other events in the batch
e_m^1 e_m^2 e_m^3 ...
push apart
$(e_v^i, e_m^j),\ j \neq i$
5 SIMILARITY MATRIX (per modality m)
Cosine similarity scaled by temperature τ
Modality embeddings (e_m)
Visual anchors (e_v)
$S_{ij}^{(m)} = \frac{1}{\tau} \cos(e_v^i, e_m^j)$
Positive pairs (diagonal): correct match (same event)
Negative pairs (off-diagonal): other events in the batch
e_v^i visual anchor (event i)
e_m^i modality embedding (same event)
e_m^i modality embeddings (other events, negatives)
$i = 1, \ldots, N$ index of visual anchor
$j = 1, \ldots, N$ index of modality embedding
d embedding dimension
τ temperature